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Thermal properties and microstructure of Roman flat window glass found in Bulgaria

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Abstract

The aim of this study is to carry out investigations in order to establish the composition and technological behaviour of the Roman glass dated from IV to VI c. For this reason the glass transition temperature (T_g) at 380 °C, the softening point at 490 °C and crystallization ability were determined. The fragment has been investigated by a classical chemical analysis, an electron probe microanalysis, differential thermal analysis, infrared spectroscopy, transmission electron microscopy and X-ray diffraction. According to the chemical composition, the fragment is soda-lime-silica glass that is typical for the production of this region. Microheterogeneities were found which are originated from the preparation method and from the corrosion caused by soil influence. As a result of additional thermal treatment the crystal phases were also established.

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1. Introduction

In the middle of III c. AD the provinces of the Roman Empire as Moesia, Thracia, Macedonia were affected by barbaric invasion and the Balkans and became an arena of social change. In these regions the material culture and art were modified. The destruction of buildings and acceptance of Christianity as an official religion led to new building [1]. Glassmaking has been developed be-

cause of the needs of flat window glass for churches and conditions of life in this period. Of the Roman period three glassmaking shops in Eskus [2], Nove [3] and Varna [4] and large numbers of glass article as well [5–10] on the territory of present Bulgaria have been discovered. The remains of Early Christian churches are preserved. They are found in major towns such as Sofia, Plovdiv, Varna, Pleven, etc.

Among the great monuments of church architecture is the basilica in Belovo. It is related to a Roman settlement, which was developed on a naturally fortified hill [11]. The building of basilica was erected at the end of V c. AD. In VI c. AD a number of Slavs passed across the Danube River towards Thessalonica and they demolished the

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basilica. Archaeological excavations organized by the National Historical Museum of Sofia have found the glass vessels and fragments of flat window glass [12]. The finds are dated from IV to VI c.

From a scientific point of view it is interesting to carry out investigations to establish the composition and technological properties of the artifacts glass. This information may determine the source of raw materials, the method of production and the corrosion processes.

2. Experimental

The object of the present paper is investigation of a fragment of flat window glass, found in the region of Belovo, Bulgaria. Visually the glass specimen studied is translucent with light-green colour, bubbles, variable thickness (from 3.0 to 3.8 mm) and a corroded surface. Rough surface with parallel furrows on one surface are observed. The chemical composition was determined by classical chemical analysis and by electron probe microanalysis (EPMA).

Thermal properties were measured by differential thermal analysis (DTA). The glass transition temperature (T_g) the softening point and the crystallization ability were determined. DTA was performed (MOM – Derivatograph OD.102, System Paulik), in the following conditions: weight of sample $g = 200$ mg, heating rate of 10 °C/min at air atmosphere and α - Al_2O_3 as a standard. The accuracy of recording of the temperature was ± 10 °C. The softening point was determined in an

electric furnace by an indenter method at heating rate of $3\text{--}4$ °C/min by indenter method. The sample was thermally treated for 24 h in furnace at 418 °C to establish the crystallization ability according to DTA data.

The microstructure of the studied glass was investigated by transmission electron microscopy (TEM) and infrared (IR) spectroscopy. The TEM (Philips EM-400) observations were made using C + Pt replica technique from the glass surface and also from fresh fractured surfaces of the samples before and after forced crystallization. IR spectrum was measured (Specord 71 IR) in nujol suspension. The accuracy of recording of the absorption maximum was ± 3 cm^{-1} . For phase characterization, X-ray diffraction (XRD) patterns were obtained (TUR 72 X-ray diffractometer) with $\text{CuK}\alpha$ radiation.

3. Results

The results of the classical chemical analysis and EPMA are presented in Table 1. For comparison, literature data are given there too. It is seen that the main components are: SiO_2 , Na_2O , CaO and Al_2O_3 , and additional components: Mn_2O_3 , CuO , Fe_2O_3 , MgO and K_2O .

DTA diagram (Fig. 1) shows an endothermic effect at 360 °C corresponding to decarbonisation and to the processes of transformation range and exothermic effect at 418 °C, which we assign to crystallization. The glass transition temperature (T_g) was determined at 380 °C by drawing tangents

Table 1

Chemical composition of investigated glass including analytical error, and comparison with literature data (Sample 1 – before forced crystallization, Sample 2 – after forced crystallization)

Oxides (wt%)	Chemical analysis	Electron probe microanalysis		Roman glass II–I c. BC [10]	Byzantine glass IV c. AD [11]	Roman glasses I–V c. AD [11]
		1	2			
SiO_2	$65.93 \pm 0.01\%$	$71.36 \pm 0.28\%$	$72.26 \pm 0.28\%$	63.63	66.22	68.10
Al_2O_3	$2.05 \pm 0.01\%$	$2.06 \pm 3.42\%$	$1.98 \pm 3.68\%$	1.95	2.62	2.52
Fe_2O_3	$1.00 \pm 0.01\%$	$0.54 \pm 8.99\%$	$0.78 \pm 7.01\%$	0.70	1.26	0.81
CaO	$5.14 \pm 0.01\%$	$7.00 \pm 0.88\%$	$7.08 \pm 0.89\%$	6.28	8.28	6.98
MgO	$0.82 \pm 0.01\%$	–	–	2.53	1.24	1.03
Mn_2O_3	$1.08 \pm 0.01\%$	$0.79 \pm 5.87\%$	$1.14 \pm 4.30\%$	0.07	0.58	0.50
CuO	$1.02 \pm 0.01\%$	–	–	0.62	–	–
K_2O	$0.57 \pm 0.01\%$	$0.37 \pm 8.01\%$	$0.38 \pm 8.27\%$	0.11	0.98	1.13
Na_2O	$23.39 \pm 0.01\%$	$17.88 \pm 1.31\%$	$16.38 \pm 1.32\%$	23.86	18.28	18.63

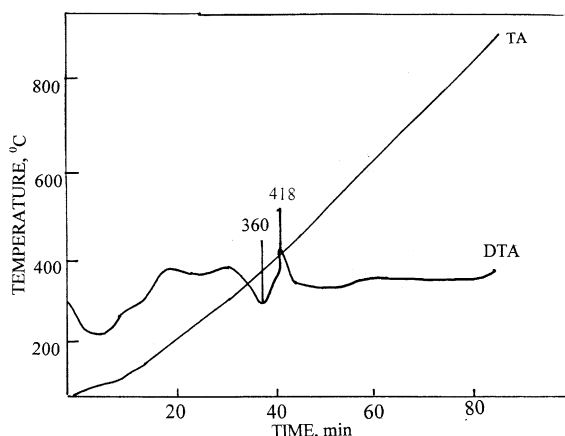


Fig. 1. DTA diagram of investigated glass.

around the diversion DTA curve. This temperature is much lower than T_g of contemporary glasses [13]. The softening point was registered at 490 °C.

The IR spectrum of the glass is presented in Fig. 2. The observed absorption bands at 1020 and 756 cm^{-1} can be assigned to the Si–O and Al–O vibrations [14].

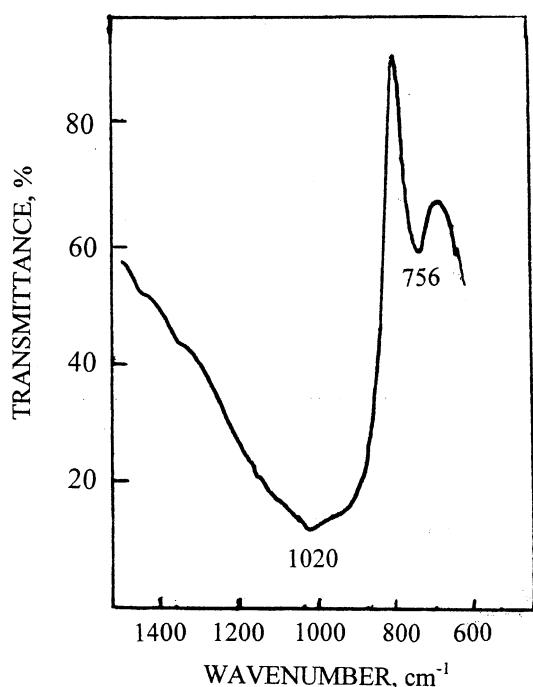


Fig. 2. IR spectrum of investigated glass.

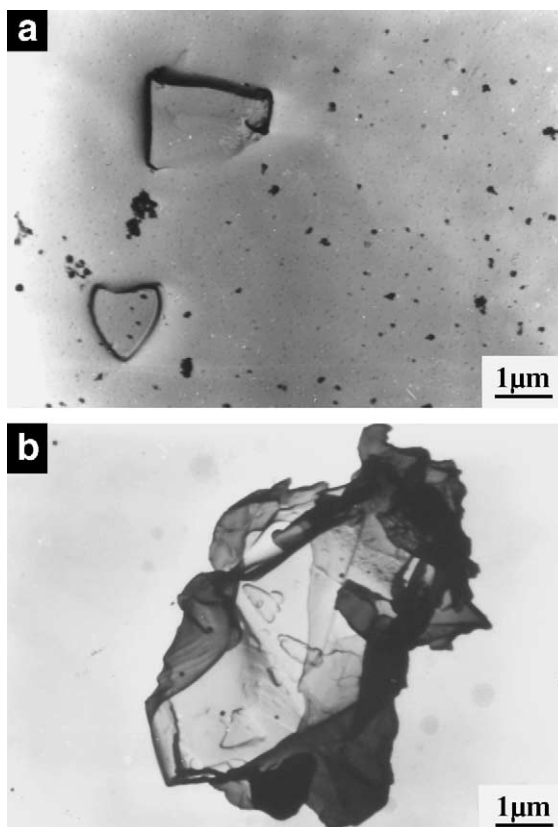


Fig. 3. Microstructure of the surface of Roman flat window glass: (a) before heat treatment; (b) after heat treatment.

The TEM observations show a typical glassy structure of the matrix with some dispersed heterogeneities in the sample surface (Fig. 3(a)) before heat treatment. After forced crystallization in the same surface, the appearance of more complex crystalline aggregates is observed (Fig. 3(b)).

According to XRD patterns of the crystallized glass, two phases were identified: nepheline – $\text{Na-AlSi}_3\text{O}_8$ ($d = 1.55, 2.33, 3.00 \text{ \AA}$) and devitrite – $\text{Na}_2\text{Ca}_3\text{Si}_6\text{O}_{16}$ ($d = 2.95, 2.88, 2.76 \text{ \AA}$) [15].

4. Discussion

On the basis of historical facts we assume that our glass is Roman wherefore we are looking for a connection between our glass and other investigated Roman glasses. According to Harden [16]

there are two different kinds of glass resulting from the method of production of the Roman glasses: transparent with smooth surfaces and translucent with different surfaces. Because of the translucent, different surfaces, presence of parallel furrows from one of its side and variable thickness as well, we presume that the investigated glass was made by roller technique [17]. For the most part the Roman glasses relate to the soda-lime-silica glass group, according to the classification of Bezborodov [18]. This type of glasses existed from XV c. BC to X c. AD in countries of the East, the Mediterranean and the Rhine area [19]. Comparison of the chemical composition of our glass with literature data shows that it is from the same group which is made by an ancient recipe with a proportion of sand:soda = 1:3 [20]. The soil medium influences on state of the glass (transparency loss) due to surface extraction of sodium ions and forming of corrosion layers of quartz, calcite and dolomite [21]. All this is seen from the EPMA data where a decrease of sodium content is registered. Endothermic effect at 360 °C obtained by DTA we associate with calcite decarbonisation. Exothermic effect at 418 °C is due to crystallization, which is confirmed by the performed experiment of thermal treatment [13]. The softening point was determined at 490 °C. The low value of the T_g can be explained with the high amount of alkaline components. A silicate type of the glass is seen also from an IR-spectrum showing absorption of Si–O and Al–O tetrahedra [14]. The observed micro-heterogeneities originated from the method used for preparation and from the corrosion caused by the soil. The small crystals of nepheline and devitrite were also established, which results from the thermal treatment.

5. Conclusions

In the present paper, a glass fragment found by archaeological excavations of a Roman object on the region of Belovo, Bulgaria is investigated. It is dated from IV to VI c. According to the chemical composition the fragment is soda-lime-glass, typical for the production of this region. Suggestion is

made that the investigated glass is made by a roller technique. The soil medium affected the state of the glass (transparency loss) due to surface extraction of sodium ions and forming of corrosion layers, which is seen from the EPMA and DTA data. The fragment has surface crystallization. The crystal phases were observed as a result of thermal treatment.

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References

- [1] Z. Vazharova, Slavs and Bulgars in IV–VI c., Bulgarian Academy of Science, Sofia, 1976, p. 405.
- [2] T. Kovacheva, Bull. Museums North-West Bulgaria 24 (1996) 35.
- [3] K. Koicheva, Archaeology 4 (1990) 36.
- [4] G. Djingov, Annales du 3^{ème} Congress des 'Journées Internationales du verre', Liège, (1964) 111.
- [5] A. Minchev, Bull. National Museum – Varna 20 (1984) 5.
- [6] A. Minchev, Bull. National Museum – Varna 24 (1988) 35.
- [7] A. Minchev, Bull. National Museum – Varna 25 (1989) 13.
- [8] A. Minchev, Bull. National Museum – Varna 26 (1990) 56.
- [9] S. Cherneva-Tilkian, Annuaire du musée archeologique – Plovdiv 8 (1997) 42.
- [10] S. Cherneva-Tilkian, Bull. Museums South Bulgaria 71 (1995) 71.
- [11] A. Hrishev, 333 ages from migration in Belovo, 'Beloprint' (1999) 32.
- [12] G. Lazov, private communication.
- [13] J. Zarzycki, Glasses and the vitreous state, in: Cambridge Solid State Science Series, Cambridge University, 1991, p. 13.
- [14] K. Mackenzie, J. Am. Ceram. Soc. 55 (1972) 68.
- [15] M. Ihara, J. Ceram. Assoc. Jpn. 92 (1984) 373.
- [16] D. Harden, Glastehn. Ber. 32-K (1959) 8.
- [17] U. Shtapova, History of Ancient Glassmaking, Moscow University, Moscow, 1983, p. 70.
- [18] M. Bezborodov, Chemistry and Technology of Ancient and Medieval Glasses, Science and Technology, Minsk, 1969, p. 153.
- [19] M. Bezborodov, The Glassmaking in Ancient Russia, Science and Technology, Minsk, 1956, p. 87, 90.
- [20] G. Djingov, Archaeology 4 (1965) 11.
- [21] W. Müller, Proc. XVI International Congress on Glass, Madrid, vol. 1 1992, p. 219.